

Solar PV Forecasting with Per-Site and Cross-Site Models on the Dryad Hong Kong Dataset

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INTRODUCTION

This report describes a supervised regression approach to the solar power forecasting task (Problem 1.3 [1]) using the Dryad Hong Kong rooftop PV dataset [2, 3]. Power and meteorological time series from 60 stations were resampled to hourly resolution and merged with Brick metadata describing each installation’s rated capacity. Eight model classes were compared using scikit-learn [4]: Persistence, Linear Regression, Ridge, Lasso, Decision Tree, Random Forest (RF), Gradient Boosting (GB), and a Multi-Layer Perceptron (MLP), spanning the main supervised regression families. Hyperparameters were selected via grid search on a chronological inner-validation split to avoid future-information leakage. Two investigative choices shaped the analysis. Evaluation is restricted to daylight hours, since nighttime power is trivially zero, which lowers error averages. Feature importances are extracted for RF and GB to check that the learned mapping rests on physically sensible predictors.

DATA AND PREPROCESSING

The dataset comprises 60 grid-connected rooftop PV stations on a subtropical university campus in Hong Kong, recording power output at 5-minute intervals over 3 years (2021–2023) [2]. A co-located weather station provides seven groups of meteorological variables: irradiance, temperature, relative humidity, wind (speed and direction), rainfall, visibility and sea-level pressure at 1-minute intervals. Installation metadata are provided through the Brick schema and include system rated capacity, panel tilt, azimuth and other site-level descriptors.

All PV and weather time series were aligned onto a common hourly timeline. For the PV data, 5-minute power measurements were averaged within each hour. For the weather data, hourly means were taken for most variables, while rainfall was summed to preserve hourly accumulation. This hourly cadence reduces noise and keeps the dataset computationally manageable (Figure 1), while remaining appropriate for a 4-hour forecasting horizon: using the raw 5-minute series would greatly increase sample count without adding much extra information. After resampling, short gaps in the weather data (up to three hours) were forward-filled, and rows with longer gaps were excluded from training rather than extrapolated.

To enable comparison across installations of very different sizes, power output was normalised by each site’s rated capacity, converted from the Brick metadata into watts. This pro-

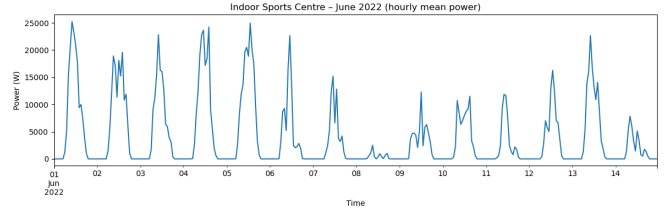


FIG. 1. Hourly mean PV power for the Indoor Sports Centre over the first two weeks of June 2022. The resampled series preserves the daily generation cycle and day-to-day weather-driven variability while removing much of the high-frequency noise present in the original 5-minute data.

duced a per-unit target, `power_pu`, so that sites with larger absolute output did not dominate the multi-site analysis. Negative normalised values were clipped to zero.

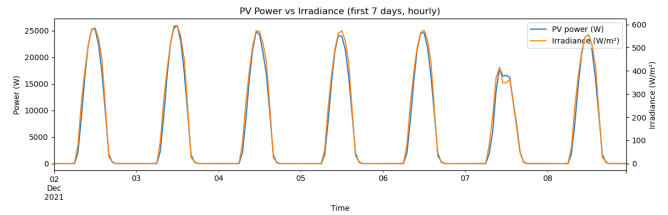


FIG. 2. Hourly PV power and irradiance over a representative 7-day period. PV output closely follows irradiance, with generation concentrated in daytime hours and negligible output at very low irradiance, motivating both the use of irradiance as a predictor and the daylight threshold applied later.

A rule-based quality filter was then applied using training-period statistics only, preventing information leakage from the test set. Sites were flagged if they had missing or invalid rated capacity, implausibly high peak output relative to rated capacity (suggesting rated-power mismatch), or a 95th-percentile output well above the fleet norm (suggesting a scale-factor error), very weak correlation between power and irradiance (inconsistent with the physical relationship shown in Figure 2), near-zero or constant output, excessive missing weather data, or insufficient training samples. This filtering left 57 sites for the per-site forecasting analysis; a further metadata-completeness restriction was applied later for the cross-site experiment only.

FEATURE ENGINEERING

Each site’s hourly time series was converted into a supervised-learning table with predictors at time t and the target set to the normalised PV output H hours ahead, $y_{\text{future}} = \text{power_pu}(t + H)$. This allows the model to use information available at the current time to forecast future generation. The headline analysis fixes $H = 4$ hours; Figure 5 compares performance across a range of horizons.

To capture short-term history, two lagged target features were added: `lag_1`, the previous hour’s per-unit output, and `lag_day`, the output 24 hours earlier. These provide information about recent persistence and the repeating daily solar cycle without using future values.

Calendar variables were encoded cyclically using sine and cosine transforms. Hour of day, day of week and month of year were each represented by $\sin/\cos$ pairs, so that values at the boundary of a cycle remain close in feature space; for example, 23:00 and 00:00 are treated as adjacent rather than numerically distant. The same approach was applied to wind direction, since 0° and 360° represent the same physical direction. This encoding is especially important for linear and scale-sensitive models, which would otherwise interpret cyclical variables as numerically distant.

A binary daylight indicator, `is_daylight`, was also included and set to 1 when irradiance exceeded 50 W/m^2 and 0 otherwise. This helps distinguish genuine daytime generation from dawn, dusk and nighttime periods where PV output is negligible and measurement noise is proportionally more important.

The final per-site feature set contained 18 predictors: two lagged power terms; six cyclical calendar features; the daylight indicator; seven scalar meteorological variables; and two cyclical wind-direction components.

MODEL SELECTION AND TRAINING

Model classes

The forecasting task was formulated as a supervised regression problem: given the feature vector available at time t , predict the normalised PV output H hours ahead. Eight model classes were compared, each described below.

The Persistence model predicts that future output will equal the most recent observed value and therefore acts as a minimum benchmark that any useful learned model should exceed. Linear Regression provides the simplest supervised baseline by fitting a linear mapping from the 18 input features to the target. Ridge (L2) and Lasso (L1) test whether regularisation improves generalisation given correlated meteorological predictors [5].

To capture nonlinear relationships, a single Decision Tree was used as a baseline. Decision trees can model threshold effects and feature interactions without requiring scaled inputs, but they are prone to high variance [6]. Therefore two other

methods were added. Random Forest reduces variance by averaging many decorrelated trees trained on bootstrap samples [7], while Gradient Boosting builds trees sequentially so that each new tree corrects the residual errors of the previous trees [8]. The two methods represent complementary responses to the bias-variance trade-off: bagging reduces variance, boosting reduces bias.

An MLP with ReLU activations and early stopping was also included as a compact neural-network benchmark [9]. RNNs and Transformers were not used because the lagged and cyclical features already supply the temporal context they would need to learn [10].

Training objective and loss function

MSE was chosen over mean absolute error (MAE) because PV forecast errors are not equally costly [6]. A handful of large errors during peak generation has a much larger operational impact than many small errors scattered across quiet periods. The quadratic penalty discourages the model from tolerating occasional extreme residuals, which MAE would allow. Standardisation to zero mean and unit variance was applied only to magnitude-sensitive models: Linear, Ridge, Lasso and the MLP. Tree-based methods were trained on unscaled inputs because their splits are based on thresholds rather than feature magnitudes.

Hyperparameter selection

Hyperparameters were selected using a two-stage chronological splitting procedure to avoid future-information leakage. For each site, the training window was divided into an inner-training subset (85%) and an inner-validation subset (15%), keeping time order intact. Each candidate setting was fitted on the inner-training data and scored on the inner-validation data, and the best setting for each model class was chosen by the mean NRMSE across the 57 filtered sites. 41 supervised configurations were tested across the seven model classes. The selected configurations were: Ridge $\alpha = 100$, Lasso $\alpha = 0.01$, Decision Tree `max_depth=6` and `min_samples_leaf=1`, Random Forest `n_estimators=200` and `max_depth=6`, Gradient Boosting `n_estimators=100` and `learning_rate=0.05`, and an MLP with a single hidden layer of 16 units and $\alpha = 0.001$. Each selected model was retrained on the full training window and evaluated once on the held-out test set. A single 85/15 split was used rather than time-series CV to keep the search computationally manageable, and a single architecture was selected per model class rather than tuned per site, providing a built-in robustness check. Figure 3 shows the inner-validation NRMSE for the winning configuration of each model class.

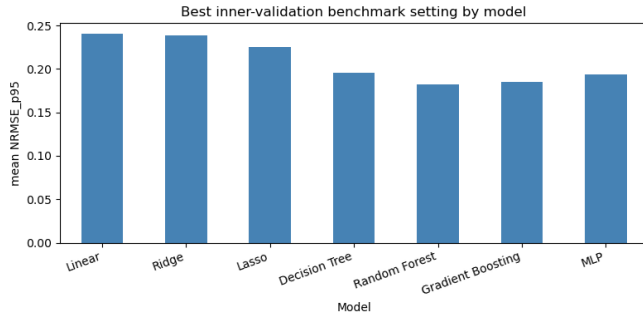


FIG. 3. Best inner-validation NRMSE for each supervised model class, averaged across the 57 filtered sites. The nonlinear models achieved the lowest validation errors (0.18–0.20), while the linear models performed slightly less well (0.22–0.24). These best-per-class configurations were then evaluated once on the held-out test set.

Evaluation protocol

Performance was assessed primarily using normalised root mean squared error (NRMSE), defined as RMSE divided by the 95th percentile of the training target. Using the training-set P_{95} as the normalisation scale provides a comparable error metric across sites of very different sizes and is a robust proxy for realistic peak output, avoiding the distortion caused by averaging in long stretches of nighttime zeros and the mismatches that can occur between nameplate rating and actual peak output (Figure 4). Normalised mean absolute error (NMAE) was also reported as a secondary metric, since it weights errors linearly rather than quadratically and is less affected by occasional large residuals.

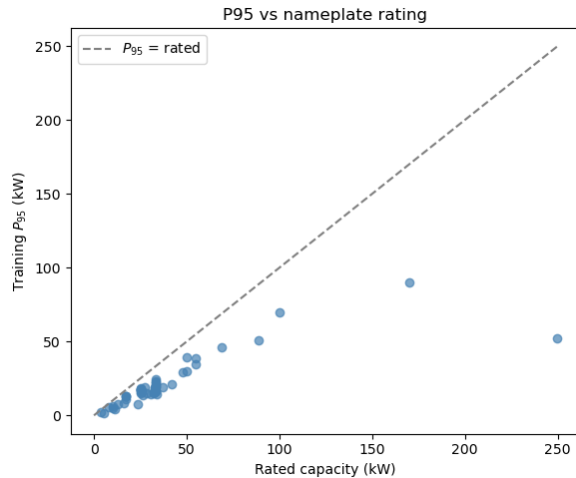


FIG. 4. Training-set 95th percentile against nameplate rating across the 57 filtered sites. Every site falls below the $P_{95} = \text{rated}$ line, with a median ratio of 0.60, meaning nameplate capacity systematically overstates observed peak output. This justifies the choice of training-set P_{95} over rated capacity as the normalisation scale for NRMSE.

Q1 — PER-SITE FORECASTING: RESULTS AND DISCUSSION

Table I presents the final test-set performance across the 57 filtered sites. Gradient Boosting achieves the lowest mean NRMSE (0.161), followed by Random Forest (0.169), the MLP (0.171) and the single Decision Tree (0.180). All supervised models substantially outperform the persistence baseline ($H = 4$ hour persistence, predicting that power in four hours equals the current value; mean NRMSE 0.393), with Gradient Boosting representing a 59% reduction in NRMSE. The win rate against persistence is 98.2% (56 of 57 sites) for every supervised model, confirming that the learned models beat persistence at nearly every site individually, not just on average. The single site where every supervised model loses to persistence is SQ567 Flexible PV, suggesting a site-specific data issue rather than a model weakness.

TABLE I. Test-set performance across 57 filtered sites, ranked by mean NRMSE. NMAE gives a broadly consistent ranking, confirming the ordering is not driven by a few extreme errors.

Model	Mean NRMSE	Mean NMAE	Win Rate (%)
Gradient Boosting	0.161	0.084	98.2
Random Forest	0.169	0.085	98.2
MLP	0.171	0.093	98.2
Decision Tree	0.180	0.090	98.2
Ridge	0.217	0.149	98.2
Linear	0.218	0.149	98.2
Lasso	0.221	0.152	98.2
Persistence	0.393	0.241	—

The two ensembles both improve on the single Decision Tree, consistent with the bias–variance trade-off introduced in Section 4. That Gradient Boosting edges out Random Forest by 0.008 NRMSE suggests that, for this deterministic-but-nonlinear problem, the marginal bias reduction from boosting outweighs the variance reduction from bagging. The clear gap between the linear models (~ 0.22) and the tree-based and neural models (~ 0.17 on the held-out test set; Table I) confirms that the relationship between features and power is substantially nonlinear, and that models capable of capturing feature interactions provide a meaningful advantage. Ridge and Lasso barely improve on plain Linear Regression: the features are not strongly correlated after cyclical encoding, so regularisation has little to correct. The MLP (single 16-unit hidden layer, $\alpha = 0.001$) is competitive but does not overtake the ensembles; larger architectures in the hyperparameter sweep did not improve on this.

Figures 5 and 6 show how forecast error scales with horizon and how predictions are distributed against truth at $H=4$. Together they confirm that the performance gap between linear and tree-based models comes from the nonlinearity of the underlying mapping, not from the linear models being poorly regularised.

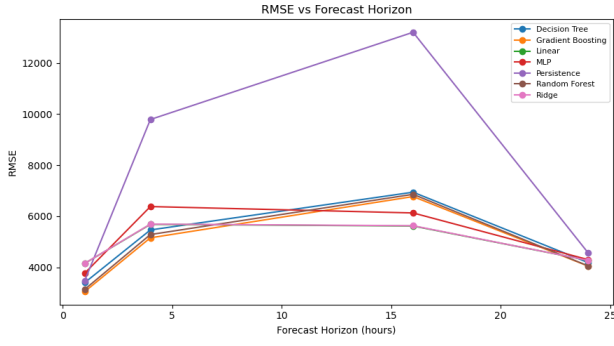


FIG. 5. RMSE against forecast horizon H for the default single site (Indoor Sports Centre). At $H=1$, all models cluster tightly and persistence is competitive: one hour ahead, predicting the next value from the current value is already a strong baseline. By $H=4$, persistence has degraded sharply (RMSE $\approx 9,800$ W) while the ensembles hold near 5,200 W, showing that the supervised models have learned genuine temporal structure rather than relying on short-term correlations. Persistence collapses further at $H=16$ (13,200 W) before recovering at $H=24$, where predicting 24 hours ahead approximates the daily cycle. The supervised models degrade gradually across all horizons and remain well below persistence in the medium-range horizon band ($H=4$ –16 hours), where persistence breaks down but learned temporal patterns still hold.

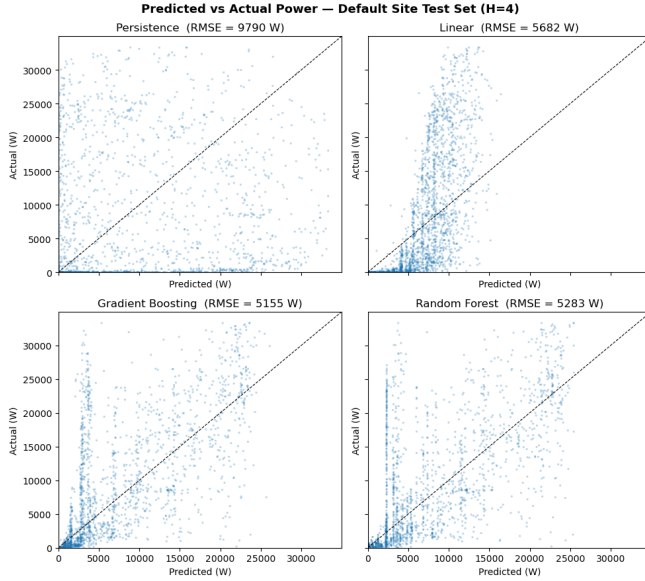


FIG. 6. Predicted against actual power at $H=4$ for the default single site (Indoor Sports Centre), four models. Persistence (top left, RMSE 9,790 W) shows broad scatter and no diagonal structure. The model has no predictive signal at this horizon. Linear regression (top right, RMSE 5,682 W) tracks the diagonal at low-to-mid output but shows a pronounced vertical stripe at low predicted values where actuals span the full range: the model cannot represent the threshold-like, zero-clipped response of PV power to irradiance. Gradient Boosting (bottom left, RMSE 5,155 W) and Random Forest (bottom right, RMSE 5,283 W) cluster around the diagonal with residual scatter concentrated at low predicted values, the characteristic signature of well-calibrated regressors that have learned the nonlinear irradiance-to-power mapping.

Daylight-only evaluation

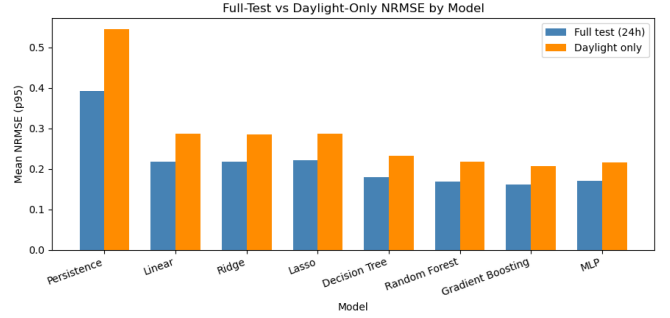


FIG. 7. Mean NRMSE under full 24-hour evaluation (blue) and daylight-only evaluation with irradiance > 50 W/m² (orange). All models degrade once nighttime zeros are removed, but not uniformly: persistence rises from 0.393 to 0.545 (+39%), the linear models by around 30–31%, and the ensembles by only 28–29% (Gradient Boosting: 0.161 to 0.207). The ranking is preserved and the 98.2% win rate against persistence holds under both splits; the gap between learned models and persistence widens when only daytime hours are scored.

Standard 24-hour evaluation includes nighttime hours where PV output is trivially zero and all models score well by default. Restricting to daylight hours (irradiance > 50 W/m²) was a deliberate investigative choice to isolate the hours where forecasting is genuinely difficult. Persistence was relying on the nighttime zeros, while the supervised models genuinely learn daytime generation patterns (Fig. 7).

Feature importance

The two ensembles produce near-identical rankings (Fig. 8), giving confidence the ranking reflects real structure rather than a quirk of one model.

Two failure modes

Figure 9 shows two representative Q1 sites that illustrate different failure modes of the tuned Gradient Boosting model. (a) is the typical-case SQ8, where an overshoot in the plotted window reflects the monsoon period being underrepresented in chronological training data, a methodological limitation. (b) is Zone A2, the largest installation, where the static meta-data schema cannot describe Zone A2’s physical configuration, a structural limitation. The second failure mode motivates the cross-site analysis in Q2.

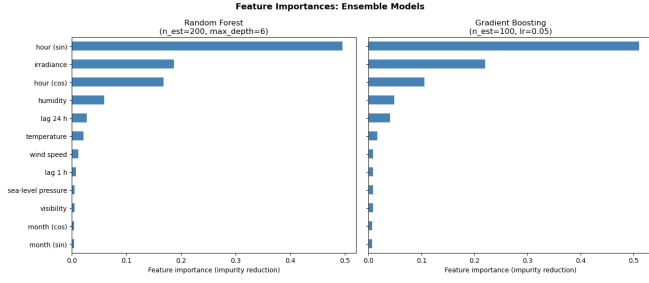


FIG. 8. Impurity-based feature importances for Random Forest (left) and Gradient Boosting (right), averaged across the 57 filtered sites. Both rank the cyclical hour encoding ($\text{hour_sin} \approx 0.50$, $\text{hour_cos} \approx 0.10\text{--}0.17$) and on-site irradiance (≈ 0.20) as the dominant predictors, together carrying over 80% of the combined importance. This is physically expected, since the daily cycle determines when generation occurs and irradiance is the primary driver of PV output. Relative humidity ranks fourth ($\approx 0.05\text{--}0.06$), plausibly because in Hong Kong’s subtropical climate humidity correlates with cloud cover and reduces the irradiance reaching the panels. The 24-hour lag ($\text{lag_day} \approx 0.03\text{--}0.04$) contributes modestly; the 1-hour lag, wind speed, sea-level pressure, rainfall and the month/day-of-week features are negligible. The low importance of the 1-hour lag is notable: persistence is essentially a lag_1 forecast, so its collapse at $H=4$ (Table I, Fig. 5) is directly attributable to the ensembles learning to ignore a feature on which persistence relies.

Q2 — CROSS-SITE GENERALISATION: RESULTS AND DISCUSSION

Forecasts are produced at the same $H=4$ hour horizon used in Q1, so the per-site and pooled results are directly comparable. The brief’s phrase ‘based on metadata about that installation’ can be read in two ways: strictly metadata-only, or as a model that uses metadata to differentiate sites alongside other shared inputs such as weather. Two variants were evaluated to address both readings: “metadata + history” uses lagged power from the held-out site at inference time, while “metadata only” removes the lag features entirely, directly answering the brief’s scenario of forecasting a new installation before any of its own operational data exist. Of the 57 sites retained after the quality filter, 40 had complete metadata for all cross-site features and were eligible for the experiment; eight of these were held out using a fixed random seed and the remaining 32 used to train a single pooled model per configuration.

TABLE II. Q2 — Cross-site NRMSE on the 8 held-out sites (mean across sites). The within-site column shows each model’s Q1 per-site result on the same 8 sites for reference. Persistence baseline (mean NRMSE = 0.396 across the held-out sites) is identical for all model rows.

Model	Metadata + History	Metadata Only	Within-Site
Gradient Boosting	0.168	0.169	0.159
Random Forest	0.186	0.187	0.167
Linear	0.228	0.231	0.212
Ridge	0.228	0.231	0.211

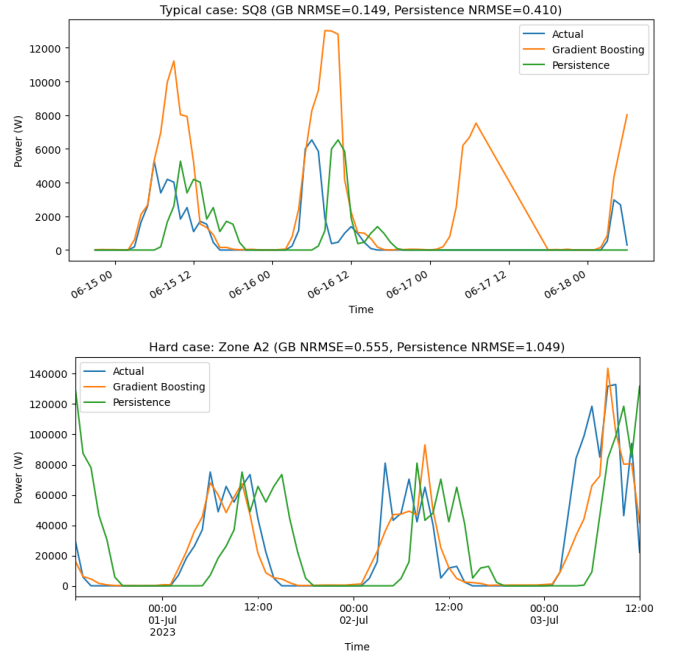


FIG. 9. Three-day test windows for two representative sites. (a) SQ8, whose site-level NRMSE (0.149) is closest to the median across the 57 filtered sites; the plotted window falls in a monsoon period (14–18 June 2023) with mean irradiance 79 W/m^2 against 213 W/m^2 for the full test set, and GB overshoots through the overcast hours. Across SQ8’s full test set, however, GB’s NRMSE is 0.149 against persistence’s 0.410, so the local window is unrepresentative. The overshoot reflects tree-based models’ known poor extrapolation beyond the training distribution, a consequence of chronological splitting. (b) Zone A2 (249.5 kW), the largest installation in the dataset; both GB (NRMSE 0.555) and persistence (1.049) struggle, reflecting a structural limitation. The static metadata schema (single tilt_deg , single azimuth_deg , plus a binary mixed_orientation flag) is too coarse to describe large installations, and Zone A2’s reported values ($\text{tilt } 0^\circ$, $\text{azimuth } 0^\circ$) appear to be placeholders rather than measurements.

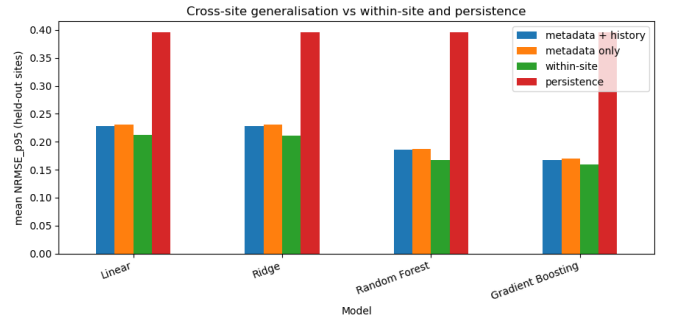


FIG. 10. Cross-site NRMSE (metadata + history and metadata only) vs within-site (trained per-site on the same held-out sites) and persistence NRMSE on the 8 held-out sites for each of the four model classes. The gap between cross-site and within-site is small for all models, and the gap between the two cross-site variants is negligible. Lagged power from an unseen site contributes almost no predictive value.

The cross-site Gradient Boosting achieves a mean NRMSE of 0.168 on the eight held-out sites, only 0.009 worse than the 0.159 it achieves when trained within-site on the same sites (Table II). A model trained on other buildings therefore forecasts an unseen installation almost as well as a model trained on that installation’s own history. This means a newly commissioned rooftop installation can be forecasted from the day it is connected, before any of its own generation data are available. Figure 11 shows this concretely for one held-out site: a three-day test window for LSK South (25 kW, excluded from training), where the pooled model tracks actual power with RMSE 3,047 W against persistence’s 6,537 W.

The “metadata + history” and “metadata only” variants differ by only 0.001–0.004 NRMSE across all model classes (Figure 10). Each site has its own baseline magnitude and operating conditions, so a single lag value carries little cross-site signal, while irradiance, the hour-of-day features, and the metadata together capture the cross-site pattern. This directly addresses the brief’s wording: the metadata-only variant, which uses no site-specific operational history at all, is within 0.001 NRMSE of the variant that does.

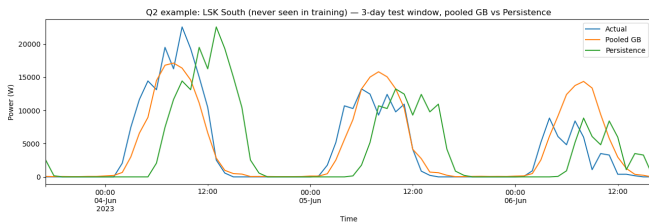


FIG. 11. Three-day test window for LSK South (25 kW), a held-out site never seen during training. The pooled Gradient Boosting model tracks actual power with RMSE 3,047 W, while persistence (RMSE 6,537 W) visibly lags the daily cycle.

A leave-one-site-out spot check at three installation sizes reveals that cross-site difficulty scales strongly with installation size (Table III), with Zone A2 (249.5 kW) degrading to GB NRMSE 0.422. This matches the Zone A2 hard case identified in Q1: the static metadata schema (single `tilt_deg`, single `azimuth_deg`, plus a binary `mixed_orientation` flag) is too coarse to describe large installations, and Zone A2’s reported values (`tilt` 0°, `azimuth` 0°) appear to be placeholders rather than measurements, leaving the pooled model with no usable physical descriptors. Richer and more reliably populated site descriptors would be needed to bring the largest installations into line with the rest of the fleet.

CONCLUSIONS

For per-site forecasting (Q1), Gradient Boosting achieved a mean NRMSE of 0.161 across 57 filtered sites, a 59% reduction over persistence and winning on 98.2% of sites. Daylight-only evaluation, a deliberate investigative choice to remove

TABLE III. Leave-one-site-out spot check using pooled Gradient Boosting (metadata + history) across three installation sizes. Cross-site NRMSE scales with installation size; persistence is shown per site.

Site	Rated power (kW)	GB NRMSE	Persistence NRMSE
UG Hall2 RF	8.0	0.125	0.307
UG Hall7	32.0	0.189	0.359
Zone A2	249.5	0.422	1.049

trivial nighttime zeros, confirmed that the supervised models genuinely learn daytime generation patterns rather than coasting on free zeros, with the gap to persistence widening under the physically meaningful metric. Feature importance analysis showed the ensembles relying on physically sensible predictors (cyclical hour encoding, on-site irradiance, relative humidity as a cloud-cover proxy) and explicitly deweighting the one-hour lag on which persistence depends, directly explaining why persistence collapses at medium-range horizons.

For cross-site generalisation (Q2), the pooled Gradient Boosting model forecasting eight entirely unseen sites achieved NRMSE 0.168, only 0.009 worse than its within-site counterpart. Metadata and weather carry almost all of the cross-site predictive power, directly answering the brief’s wording. The practical implication is that a newly installed rooftop system can be forecasted from day one, before any operational history exists, with the caveat that difficulty scales with installation size and that the static metadata schema is too coarse for large installations whose reported values appear to be placeholders.

USE OF GENERATIVE AI

ChatGPT (OpenAI, GPT-5.4) and Claude (Anthropic, Opus 4.6) were used as coding assistants for data-loading, cross-site experiment infrastructure, additional model classes, and plotting cells, with inline attribution in the notebook; see the notebook’s Use of AI section for full detail. The problem framing, investigative choices, failure-mode diagnoses, and result interpretations are my own.

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